INCH-POUND

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MILITARY SPECIFICATION

AMMONIUM PERCHLORATE, CONDITIONED (WITH REDUCED ALKALI METAL CONTENT)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of ammonium perchlorate (AP) conditioned with tribasic calcium phosphate (TCP), and with a reduced level of alkali metal content, referred to herein as the AP.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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SPECIFICATIONS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-D-3464 Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001).

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

ASTM E 11 Standard Specification for Wire-Cloth Sieves for Testing Purposes (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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UNITED STATES PHARMACOPEIAL CONVENTION, INC

National Formulary Tribasic Calcium Phosphate

(Application for copies should be addressed to the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the AP shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Material.** The AP shall be conditioned with 0.2 percent, nominal, tribasic calcium phosphate, an anti-caking agent, conforming to National Formulary Tribasic Calcium Phosphate (see 6.2).

3.3 **Characteristics.** The AP shall be a granular crystalline solid consisting essentially of whole crystals with rounded edges and corners.

3.4 **Chemical and physical properties.** Chemical and physical properties of the AP shall conform to the requirements specified in table I.

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TABLE I. Chemical and physical properties.

Property	Percent by Weight	
	Minimum	Maximum
Moisture:		
Total	---	0.04
Internal	---	0.03
External	---	0.01
Acid insolubles	---	0.04
pH of aqueous solution 1/	5.0	6.5
Chloride, as ammonium chloride	---	0.155
Sulfated ash, as sodium perchlorate	---	0.9
Chlorate, as ammonium chlorate	---	0.02
Iron, as ferric oxide	---	0.0036
Perchlorate, as ammonium perchlorate	98.3	---
Phosphate, as TCP	0.10	0.25
Potassium content	see 3.4.1	
Sodium content	see 3.4.1	
Particle size distribution 2/		
Cumulative percent retained on:		
No. 40 sieve (425 µm)	0	3
No. 50 sieve (300 µm)	0	11
No. 70 sieve (212 µm)	7	38
No. 100 sieve (150 µm)	48	86
No. 140 sieve (106 µm)	85	99
No. 200 sieve (75 µm)	95	100

1/ Limit represents acidity and not a percent by weight.
2/ U. S. Standard sieve as defined by ASTM E 11.

3.4.1 **Potassium and sodium content.** The total allowable level of potassium and sodium shall be determined by the following equation:

ppm potassium + 1/10 ppm sodium $\leq$ 20 ppm

Where: ppm = parts per million

3.5 **Shelf life.** The AP specified herein shall meet the requirements of this specification for a period of 18 months from the date of quality

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conformance inspection (see 4.4) when packaged as specified in section 5 and stored at ambient conditions.

3.5.1 **Shelf life extension.** The shelf life of an individual lot may be extended for additional 18-month periods provided the AP, upon retest, conforms to the total moisture, external moisture, internal moisture and particle size distribution requirements.

3.6 Safety

3.6.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.6.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.7 **Workmanship.** Workmanship shall be such that the AP shall be free-flowing, uniform in appearance, and free from visible contamination or any other defects which would render the AP unsuitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for

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review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspection (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. Quality conformance inspection.

Examination and tests	Requirement	Test method
Visual examination	3.2, 3.3, 3.7, and section 5	4.5.1
Total moisture	3.4	4.5.2
External moisture	3.4	4.5.3
Internal moisture	3.4	4.5.4
Acid insolubles	3.4	4.5.5
Hydrogen ion (pH)	3.4	4.5.6
Chloride	3.4	4.5.7
Sulfated ash	3.4	4.5.8
Chlorate	3.4	4.5.9
Iron	3.4	4.5.10
Perchlorate	3.4	4.5.11
Phosphate	3.4	4.5.12
Potassium content	3.4.1	4.5.13
Sodium content	3.4.1	4.5.14
Particle size distribution	3.4	4.5.15

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** A lot shall consist of all the AP subject to inspection at one time from one of the following:

a. A single batch (see 6.5.1).

b. A continuous process (see 6.5.2).

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c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 **Selection of sample containers.** Sample containers of the AP shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.15. Sample material shall be selected as specified in 4.4.1.3.

TABLE III. Sampling for inspection.

Number of containers in the lot	Number of sample containers to be selected for inspection
1 - 3	All
4 - 15	3
16 - 25	4
26 - 40	5
41 - 65	7
66 - 110	10
111 - 180	15
181 - 300	20
301 - 500	25
501 - 800	30
801 - 1300	35

4.4.1.3 **Sampling material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.15. For a lot size of less than three containers, three single samples shall be taken from each container and blended and riffled into three composite samples for testing.

4.4.1.4 **Contractor source sampling.** Unless otherwise specified in the contract, material stream sampling shall be performed while loading drums or bulk containers. The samples shall be shipped to the contracting activity or other testing laboratory for testing as indicated in the contract.

4.4.1.4.1 **Drum containers.** As each drum container is loaded, a minimum of one sample (1 pound minimum size) shall be obtained. The samples shall be obtained at the beginning, at equal intervals, and at the end of the loading operation.

4.4.1.4.2 **Bulk containers.** As each bulk container is loaded, a minimum of five samples (5 pound minimum size per sample) shall be obtained. The samples shall be obtained at the beginning, at equal intervals, and at the end of the loading operation.

4.4.2 **Selection of samples.**

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4.4.2.1 **Drum containers.** Sampling from drum containers shall consist of obtaining a minimum of four samples from each drum selected for sampling (see 4.4.1.2). The samples shall be prepared for testing in accordance with 4.4.3.

4.4.2.2 **Bulk containers.** Sampling from bulk containers shall consist of obtaining a minimum of two samples from each bulk container selected for sampling (see 4.4.1.2). The samples shall be prepared for testing in accordance with 4.4.3.

4.4.3 **Preparation of samples for testing.** Arrange the samples according to the numerical sequence of container sample selection (see 4.4.1.2). Divide the samples into three groups for compositing. The number of samples in each group need not be equal. Blend and mix by a riffling process, all the samples from the first third of the container samples until the composite is uniform. Obtain a sufficient quantity of AP to perform the tests specified in table II. Blend and mix the remaining two groups of samples in a similar manner and obtain the necessary quantities for testing. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

4.4.3.1 **Testing of samples.** A single determination of each applicable quality characteristic listed in table II shall be performed on each of the three composite samples of 4.4.3 independently. The results of each test shall meet the requirements of table I (see 6.3).

4.5 **Test methods.** The following test methods shall be used. Unless otherwise specified in the test description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** All samples shall be visually examined for conformance to 3.3, 3.7 and section 5. Contractor certification verifying conformance to 3.2 shall be required (see 6.2).

4.5.2 **Total moisture.** Total moisture shall be determined as follows:

a. Weigh a 15-gram (g) sample to the nearest 0.1 g.

b. Dissolve the sample in 100 ± 20 milliliters (ml) of 3:1 pyridine-methanol carrier solution which has been previously neutralized with Karl Fischer (KF) reagent to a 30-second endpoint.

c. Using an automatic titrimeter, titrate the moisture with dilute standardized KF reagent (prepared by diluting commercial KF reagent with pyridine or KF reagent diluent to a water equivalent of 1.8 to 2.3 milligrams (mg) of distilled water per ml of KF reagent).

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d. Consider the determination complete when an endpoint is reached which is stable for 30 seconds.

e. Calculate the total moisture by the following equation:

$$\text{Percent moisture} = \frac{VE}{10W}$$

Where: V = volume of standardized KF reagent, ml

E = water equivalent of standardized KF reagent, mg/ml

W = weight of sample, g

10 = unit/percentage conversion factor

4.5.3 **External moisture.** External moisture shall be determined as follows:

a. Determine external moisture in the same manner as outlined for total moisture (4.5.2), except as follows:

(1) Use a 20-g sample weighed to the nearest 0.1 g.

(2) Use 100 ± 20 ml of carrier solution previously saturated with ammonium perchlorate and subsequently neutralized with KF reagent to a 30-second endpoint.

b. Record external moisture to the nearest 0.001 percent.

4.5.4 **Internal moisture.** Internal moisture shall be determined as follows:

a. Subtract the surface moisture from the total moisture.

b. Record the difference as internal moisture.

c. Record internal moisture to the nearest 0.001 percent.

4.5.5 **Acid insolubles.** Acid insolubles shall be determined as follows.

a. Weigh a 25-g sample to the nearest 0.1 g.

b. Dissolve the sample in 250 ± 50 ml of distilled water and add 5 ml of 10 percent hydrochloric acid (HCl).

c. Filter the solution through a tared medium-porosity Selas crucible and wash with distilled water.

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d. Dry the crucible for 90 ± 10 minutes at 105 +5, -0 degrees Celsius (°C), cool to room temperature and weigh to the nearest 0.1 mg.

e. Calculate the acid insolubles by the following equation:

$$\text{Percent acid insolubles} = \frac{(A - B)}{W} 100$$

Where: W = sample weight, g

B = crucible weight, g

A = weight of crucible plus residue, g

f. Record percent acid insolubles to the nearest 0.001 percent.

4.5.6 **Hydrogen ion (pH).** pH shall be determined as follows:

a. Weigh a 20-g sample to the nearest 0.1 g.

b. Dissolve the sample in 50 ml of neutral, distilled water.

c. Measure the pH of the solution with a pH meter previously calibrated with buffer solutions.

d. Record pH to the nearest 0.1 unit.

4.5.7 **Chloride.** Chloride shall be determined, using the Volhard method, as follows:

a. Weigh a 10-g sample to the nearest 0.1 mg.

b. Dissolve the sample in 100 ± 20 ml of distilled water and 5 ml of 6 Normal (N) nitric acid.

c. Add 5.0 ml of standard 0.05 N silver nitrate plus 2 ml of nitrobenzene and 2 ml of 40 percent ferric alum indicator.

d. Back titrate the solution with standard 0.05 N potassium thiocyanate.

e. Calculate percent chloride by the following equation:

$$\text{Percent chloride as ammonium chloride} = \frac{(AB-CD)}{W} 5.35$$

Where: A = volume of standard silver nitrate, ml

B = normality of standard silver nitrate

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C = volume of standard potassium thiocyanate, ml

D = normality of standard potassium thiocyanate

W = sample weight, g

5.35 = factor for percent ammonium chloride in this reaction

f. Record chloride to the nearest 0.001 percent.

4.5.8 **Sulfated ash.** Sulfated ash shall be determined as follows:

a. Weigh a 10-g sample to the nearest 0.1 mg into a tared silica crucible. Wet the sample with 50 percent sulfuric acid, place the cover on the crucible and heat over a burner to initiate decomposition.

b. After all reaction has ceased, heat the residue to evolve all fumes and place in a muffle furnace for a minimum of 10 minutes at 800 ± 50°C.

c. Remove from the muffle furnace, cool to room temperature in a desiccator, weigh to the nearest 0.1 mg and save the residue for the iron test (see 4.5.10).

d. Calculate the percent sulfated ash as sodium perchlorate by the following equation:

$$\text{Percent sulfated ash} = \frac{172\ (B-C)}{A-C}$$

Where: A = sample weight plus crucible weight, g

B = residue weight plus crucible weight, g

C = crucible weight, g

172 = gravimetric factor for conversion of sodium sulfate to percent sodium perchlorate

e. Record sulfated ash, as sodium perchlorate, to the nearest 0.01 percent.

4.5.9 **Chlorate.** Chlorate shall be determined as follows:

a. Weigh a 10-g sample to the nearest 0.01 g into an Erlenmeyer flask and add 75 ± 15 ml of distilled water.

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b. Add 10.00 ml of 1 percent ferrous sulfate solution, using a pipette or buret, and add 10 ml of mixed 15 percent sulfuric acid and 10 percent orthophosphoric acid solution.

c. While stirring, boil for 5 minutes, then stopper, and cool to room temperature.

d. Titrate with 0.01 N potassium permanganate to the first faint-pink endpoint.

e. Carry at least two reagent blanks throughout the entire procedure.

f. Calculate percent chlorate, as ammonium chlorate, by the following equation:

$$\text{Percent chlorate} = \frac{(BN-SN)\ 1.6917}{W}$$

Where: B = ml of potassium permanganate required to titrate blank

S = ml of potassium permanganate required to titrate sample

N = normality of potassium permanganate

W = sample weight, g

1.6917 = factor for percent ammonium chlorate in this reaction

g. Record chlorate as ammonium chlorate to the nearest 0.001 percent.

4.5.10 Iron. Iron shall be determined, using the sulfated ash residue from 4.5.8, as follows:

4.5.10.1 Preparation of standard iron curve:

a. Weigh 0.050 g (to the nearest 0.1 mg) of bright reagent grade iron wire. Place the wire into a 1000-ml volumetric flask. Add 2 ml of concentrated hydrochloric acid and a few ml of distilled water. Heat until solution is complete. Cool and dilute to volume and mix well. Each ml of the solution now contains 50 micrograms of iron.

b. Measure various amounts of the solution into 100-ml volumetric flasks with a pipette.

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c. Add 10.0 ml of 0.25 percent o-phenanthroline solution, 1.0 ml of 25 percent sodium citrate solution, mix well and dilute to volume. Allow to stand for at least 30 minutes at not less than 25^0C.

d. The color is developed and read. Plot the absorbency versus the micrograms of iron. The best line is fitted to the points.

4.5.10.2 **Procedure.**

a. Add 1 ml of concentrated nitric acid plus 5 to 10 ml of hydrochloric acid to the residue from 4.5.8 and gently boil down (avoid spattering) until the sample is just dry.

b. Cool the residue, moisten with 2 drops of hydrochloric acid and quantitatively transfer to a 100-ml volumetric flask. Add 1.0 ml of 10 percent hydroxylamine hydrochloride solution, mix and allow to stand for 15 minutes.

c. Add 10.0 ml of 0.25 percent o-phenanthroline solution, 1.0 ml of 25 percent sodium citrate solution, mix well and dilute to volume. Allow to stand for at least 30 minutes at not less than 25^0C.

d. Using a blank for comparison, determine the absorbency of the sample with a spectrophotometer set at 510 nanometers (nm) wave length.

e. Calculate the iron content from a standard curve.

f. Record percent iron to the nearest 0.0001 percent.

4.5.11 **Perchlorate.** Perchlorate shall be determined as follows:

4.5.11.1 **Special reagent.** Special reagent, sodium methylate, required for the perchlorate analysis, shall be prepared as follows:

a. Dissolve 6 g of metallic sodium in 200 ml of anhydrous methanol.

b. Cool the solution to prevent a violent reaction, and protect the solution from atmospheric carbon dioxide and moisture.

c. Bring the solution to 2000 ml by adding 300 ml of methanol and 1500 ml of benzene.

d. Standardize the solution against benzoic acid.

4.5.11.2 **Procedure.**

a. Weigh a 0.7-g sample to the nearest 0.1 mg.

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b. Dissolve the sample in 50 ml each of methyl ethyl ketone (MEK) conforming to ASTM D 740 and N-N-dimethylformamide.

c. Continuously purge the flask with nitrogen to prevent reaction with atmospheric carbon dioxide.

d. Add 10 drops of 0.2 percent thymol blue indicator solution (dissolved in methanol) and titrate to a pale green endpoint with standard 0.1N sodium methylate.

e. Carry a reagent blank throughout the entire process.

f. Calculate perchlorate, as ammonium perchlorate, by the equation:

$$\text{Percent ammonium perchlorate} = \frac{11.75\,(A-B)N - (2.20C + 1.16D)}{W}$$

Where: A = sample titration, ml

B = blank titration, ml

N = normality of standard sodium methylate

C = ammonium chloride in sample, percent (from 4.5.7)

D = ammonium chlorate in sample, percent (from 4.5.9)

W = sample weight, g

2.20 = gravimetric factor for conversion of ammonium chloride to ammonium perchlorate

1.16 = gravimetric factor for conversion of ammonium chlorate to ammonium perchlorate

11.75 = factor for percent ammonium perchlorate in this reaction

g. Record perchlorate as ammonium perchlorate to the nearest 0.1 percent.

4.5.12 **Phosphate as TCP.** Phosphate, as TCP, shall be determined as follows:

a. Weigh a 25-g sample to the nearest 0.01 g into a 250-ml volumetric flask.

b. Add 50 ml of distilled water, 25 ml of perchloric acid (70 to 72 percent), dissolve the sample, and dilute to volume with distilled water.

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c. Pipet a 25-ml aliquot into a 100-ml volumetric flask. Add 20 ml of water, 15 ml of perchloric acid (70 to 72 percent), and cool to room temperature. Add 10.0 ml of 0.25 percent ammonium vanadate solution (prepared by dissolving 2.350 g of ammonium vanadate in 400 ml of hot distilled water, 14 ml of perchloric acid (70 to 72 percent) cooling and diluting to 1 liter) to the solution.

d. Add 10.0 ml of 10 percent ammonium molybdate solution (104 g of ammonium molybdate, 81-percent molybdenum trioxide dissolved in distilled water to make 1 liter) and dilute to volume with distilled water. Mix well and allow to stand for at least 30 minutes at room temperature.

e. With a blank for comparison, determine the absorbency of each solution at a wave length of 470 nm using a spectrophotometer with matched 1-centimeter silica cells.

f. Determine the phosphate present from a standard curve.

g. Record phosphate, as TCP, to the nearest 0.01 percent.

4.5.13 **Potassium content.** Potassium content shall be determined in accordance with the following procedures:

4.5.13.1 **Reagents.**

a. Standard solution is prepared by dissolving 0.1907 g reagent potassium chloride in distilled water and diluting to 1 liter (1 ml is equal to 0.1 mg potassium).

b. Dilute standard is prepared by transferring 1-, 2-, and 3-ml aliquots of the standard solution into 100-ml volumetric flasks. Dilute to volume with distilled water.

4.5.13.2 **Procedure.**

a. Weigh a 1-g sample (to the nearest mg) of ammonium perchlorate and transfer to a clean 100-ml volumetric flask. Add 1 ml of 1 N hydrochloric acid and dilute to volume with distilled water.

b. Using a Beckman DU spectrophotometer (or equivalent), adjust the instrument in accordance with manufacturer's instructions.

c. Determine percent transmittance for each of the solutions (sample and standard) after the instrument is set at 100 percent for the high standard. Use distilled water for blank. Make the readings decreasing in intensity. Repeat. Average the two readings for each solution and subtract the blank reading.

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d. Prepare a plot of percent transmittance versus mg potassium.

e. Calculate the ppm of potassium in ammonium perchlorate as follows:

$$\text{ppm K} = \frac{\text{mg} \times 1000}{\text{w}}$$

Where: mg = mg of potassium (from plot)

w = weight of sample, g

1000 = unit conversion factor

f. Record potassium to the nearest ppm.

4.5.14 **Sodium content.** Sodium content shall be determined in accordance with the following procedures:

4.5.14.1 **Reagents.**

a. Standard solution is prepared by dissolving 0.2542 g reagent sodium chloride in distilled water and diluting to 1 liter (1 ml is equal to 0.1 mg sodium).

b. Dilute standard is prepared by transferring 1-, 2-, and 3-ml aliquots of the standard solution into 100-ml volumetric flasks. Bring to mark with distilled water.

4.5.14.2 **Procedure.**

a. Weigh a 1-g sample (to the nearest mg) of ammonium perchlorate and transfer to a clean 100-ml volumetric flask. Add 1 ml of 1 N hydrochloric acid and bring to mark with distilled water.

b. Using a Beckman DU spectrophotometer (or equivalent), adjust the instrument in accordance with manufacturer's instructions.

c. Determine percent transmittance for each of the solutions (sample and standard) after the instrument is set at 100 percent for the high standard. Use distilled water for blank. Make the readings decreasing in intensity. Repeat. Average the two readings for each solution and subtract the blank reading.

d. Prepare a plot of percent transmittance versus mg sodium.

e. Calculate the ppm of sodium in ammonium perchlorate as follows:

$$\text{ppm sodium} = \frac{\text{mg} \times 1000}{\text{w}}$$

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Where: mg = mg of sodium (from plot)

w = weight of sample, g

1000 = unit conversion factor

f. Record sodium to nearest ppm.

4.5.15 **Particle size distribution.** Particle size distribution, using the Ro-Tap method, shall be determined in accordance with the following procedure:

a. Riffle a sample to a range between 60 and 120 g.

b. Weigh the sample to the nearest 0.1 g and shake for 13.0 ± 0.5 minutes on sieves (see table I) which have been calibrated with National Institute of Standards and Technology glass beads.

c. Record the results as cumulative percent retained on each sieve and use the cumulative weight in making the calculations.

d. Correct the results to nominal by using Log Probability paper.

e. Correct the results for each sieve (see table I) to the nearest 0.1 percent.

4.6 **Inspection of packaging.** The AP containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The AP shall be packaged in accordance with standard commercial practice (see 6.5.3) to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.1.1.1 **Bulk containers.** When the AP is shipped in bulk containers or drums, the closure shall be properly secured. A metal seal shall be affixed in a manner to give positive proof that the container has not been opened either by opening the clamp or by removal of the nut from the clamp tightening bolt. The bolt shall be tightened to a torque of 80 to 100 inch pounds.

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NOTE: The clamp must be tapped during the torque measurement to overcome the friction between the clamp and the closure permitting a valid torque value.

5.1.1.2 **Desiccant.** The AP shall be packed with units of desiccant conforming to MIL-D-3464. The units of desiccant, enclosed in container, shall be placed on top of the AP. Quantities of desiccant required shall be as follows:

a. AP shipped in drums shall be packed with a minimum of 32 units of desiccant per drum.

b. The number of units of desiccant to be packed with AP shipped in bulk containers shall be as specified in the contract or purchase order, but shall be not less than 600 units per container.

5.1.2 **Packaging of source samples for delivery.** Each source sample obtained in 4.4.1.4.1 and 4.4.1.4.2 shall be enclosed in a plastic bag and labeled with the lot number and container or drum numbers which it represents. The samples shall be packed in 30-gallon drums lined with clean polyethylene bags to a maximum weight of 260 pounds per drum.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, external packing shall be marked and labeled for Propellant Explosive Class B in accordance with 49 CFR 171-178 and shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture (exterior shipping container only).

d. Storage conditions.

e. Toxicity and safety warning.

f. Shelf life and inspection/test date for extension (see 3.5). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 18 MONTHS PROVIDED THE AMMONIUM PERCHLORATE, UPON RETEST, CONFORMS TO THE TOTAL MOISTURE, EXTERNAL MOISTURE,

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INTERNAL MOISTURE AND PARTICLE SIZE DISTRIBUTION REQUIREMENTS OF MIL-A-82830."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The ammonium perchlorate specified herein is intended for use as a component in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required (see 3.1).

d. Certificate of Conformance (FAR Clause 52.246-15) required (see 3.2 and 4.5.1).

e. Test facility for first article inspection (see 4.3).

f. Material Safety Data Sheets required (see 3.6.2).

g. Whether the contractor shall perform acceptance testing and test facility (see 4.1).

h. Whether source sampling is required and test facility (see 4.4.1.4).

i. Packaging if other than as specified in 5.1.

j. Number of desiccant units required (see 5.1.1.2).

k. Special marking if other than as specified in 5.2.2.

l. Safety precautions (see 3.6 and 6.7).

m. Size of containers (see 5.1.1.1).

6.3 **Consideration of data requirements.** The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure than only essential data are requested/provided and that DID's are tailored to reflect the requirements of

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the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.3.1	DI-T-2072	Reports, test	Contractor format

The above DID was cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 **First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the AP sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 **Definitions.**

6.5.1 **Batch.** A batch is that amount of AP made as one unit in the manufacturing process.

6.5.2 **Continuous process.** A continuous process is an uninterrupted process of manufacturing from which, in one unbroken flow, the AP is accumulated for a lot.

6.5.3 **Standard commercial practice.** Standard commercial practice is one of the following:

a. AP loaded into 30-gallon drums which are lined with new, close fitting polyethylene bags. The bags are closed by tying.

b. AP loaded into approved bulk containers.

6.6 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal

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Acquisition Regulation (FAR) Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DoD 4145.26M) are applicable and should be specified in the contract as required by the DoD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.8 **Hazard notice.** The material described herein is flammable or explosive or both. Consequently, it presents hazards in manufacturing, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.9 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mark 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project No. 6810-N059)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82830	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
AMMONIUM PERCHLORATE, CONDITIONED, (WITH REDUCED ALKALI METAL CONTENT)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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